JASON ZHI LIANG, Ph.D.

Principal Research Scientist | Evolutionary AI | Self-Improving Agentic Systems

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Research Summary

Principal Research Scientist at Cognizant AI Lab, leading work on **automated AI research**, **open-ended learning**, and **self-improving agentic systems**, built on a decade of large-scale neuroevolution and evolutionary optimization for deep learning. My focus: automating the AI-research loop, from designing architectures and training recipes to knowledge synthesis, with minimal human intervention, guided by two convictions: a system's founding values compound over its lifetime, and open-endedness and creativity must continue at inference time, not just during training.

My work spans three layers of the AI stack: **CoDeepNEAT** (evolutionary neural architecture search, GECCO 2018), **EPBT** (loss-function and hyperparameter evolution, GECCO 2021), and **Caesar** (Deep Research agent, ICML 2026 Human-AI Co-Creativity Workshop), with the same principle recurring at each level: evolve and select architectures and training recipes below, then critique and refine an agent's own reasoning above. Caesar's adversarial refinement over a dynamic knowledge graph outperforms the strongest of three frontier deep-research agents by 13–23% on a blinded, LLM-judged creative-synthesis benchmark. Named inventor on 5 issued U.S. patents in evolutionary AI and neural architecture search.

Technical Expertise

- **Evolutionary optimization & architecture search:** neuroevolution, neural architecture search, population-based training, multi-task learning, evolutionary loss/hyperparameter optimization.
- **Agentic & self-improving systems:** deep-research agents, multi-agent orchestration, inference-time exploration, values-guided self-improvement, LLM-as-judge evaluation.
- **Distributed systems for search:** custom asynchronous scheduling and parallel evaluation of evolutionary and population-based search across GPU clusters.

Professional Experience

Cognizant AI Lab | Principal Research Scientist

San Francisco, CA | Dec 2018 – Present | Lead Researcher driving the roadmap for the lab's automated AI research effort.

1. Autonomous Self-Improvement Systems

- **Values-Based Self-Improvement Framework:** Autonomous self-improvement system, built on Claude Code and organized around three principles: (1) **values-first design**, a system's founding values as the load-bearing axis for compounding self-improvement; (2) **safety-by-design**, an immutable safety floor the agent cannot rewrite; (3) **verified acceptance**, where a self-improvement is kept only after it holds up on a held-out test and passes an independent check. This is early-stage research into how founding values shape long-horizon self-improvement.

2. Caesar: Creative Reasoning & Knowledge Synthesis

- **Automated Research via Inference-Time Self-Refinement:** Designed and implemented "Caesar," an autonomous Deep Research agent. Scaled inference-time compute using an adversarial refinement loop that actively critiques internal drafts, generates orthogonal queries to target weaknesses, and consolidates findings via a generative merge. **Outperforms the strongest of three frontier deep-research agents by 13–23% on a blinded, LLM-judged creative-synthesis benchmark (Cliff's $\delta \geq 0.76$).**
- **Graph-Based Associative Reasoning:** Replaced static RAG retrieval with a "Perceive-Think-Act" loop over a dynamic knowledge graph, enabling discovery of non-obvious cross-disciplinary connections that linear retrieval tends to miss.
- **Open-Ended Exploration Policy:** Designed an active information-foraging policy that detects stagnation in memory and graph structures, autonomously executing strategic backtracking to maximize information gain over long horizons. **Ablation: reducing exploration budget from 1000 to 250 iterations measurably degrades long-form answer quality, validating that deep topological traversal is essential for creative synthesis.**

3. Multi-Agent Systems & Large-Scale Evolutionary Optimization

- **Multi-Agent Code Generation:** Developed a **Finite State Machine (FSM)** framework to coordinate teams of specialized coding agents. Applied evolutionary algorithms to optimize the *composition* of these teams, evaluated on SciCode and HumanEval+ code-generation benchmarks.
- **Evolutionary Population-Based Training (EPBT):** Pioneered a population-based approach that evolves loss functions and hyperparameters jointly with weights via population-based weight sharing. **Outperforms the PBT baseline by 1.3pp on CIFAR-10 ResNet-32 (92.79% vs 91.53%) while exploring 520 candidate loss functions at 12.5× lower search cost than independent training** (GECCO 2021; U.S. patent pending).
- **Code Repository Analysis:** Created "Code Archaeologist," an agentic system capable of reasoning over repository histories (via Git/LFS) to detect architectural patterns and technical debt.

Sentient Technologies | Research Scientist (prev. Research Intern)

San Francisco, CA | Dec 2015 – Nov 2018

- **Distributed Evolutionary Architecture Search:** Scaled **CoDeepNEAT** (evolutionary neural architecture search, developed during my UT Austin PhD) to GPU clusters. Validated that population-based search scales with parallel compute; established asynchronous evolutionary optimization at scale and forms the basis for multiple issued U.S. patents.

- **Multi-Task Learning:** Achieved then state-of-the-art results on the **20-task Omniglot multi-task learning benchmark (88% accuracy, +21pp over Soft Ordering at 67%)** by evolving modular topologies with cross-task weight sharing.
- **Production Deployment:** Collaborated with Sentient's product team to productionize CoDeepNEAT for image captioning.

The University of Texas at Austin | Research Assistant / Ph.D. Candidate

Austin, TX | Sep 2013 – Dec 2018

- **Bilevel Meta-Evolution for Control:** Developed **MEA (Meta-Evolutionary Algorithm)**, a bilevel framework that evolves the hyperparameters of neuroevolution; the meta-level objective is non-convex and non-differentiable, so gradient methods do not apply there. Applied to continuous-control tasks including helicopter hovering and double pole balancing.

Selected Publications & Patents

Full publication list available on Google Scholar

Caesar: Deep Agentic Web Exploration for Creative Answer Synthesis (2026) *Jason Liang, Elliot Meyerson, Risto Miikkulainen*. arXiv:2604.20855. Presented at the **Human-AI Co-Creativity Workshop at ICML 2026**.

- Autonomous research agent that scales inference-time compute through an adversarial refinement loop over a dynamic knowledge graph, outperforming leading Deep Research products on a blinded, LLM-judged creative-synthesis benchmark.

Asynchronous Evolution of Deep Neural Network Architectures (2024) *Jason Liang, Hormoz Shahrzad, Risto Miikkulainen*, Applied Soft Computing, Vol. 152.

- Techniques for asynchronous population-based optimization at scale, essential for efficient training on massive distributed clusters.

Regularized Evolutionary Population-Based Training (2021) *Jason Liang, Santiago Gonzalez, Hormoz Shahrzad, and Risto Miikkulainen*, GECCO 2021.

- Method to dynamically optimize loss functions during training, improving convergence speed and generalization.

Evolutionary Neural AutoML for Deep Learning (2019) *Jason Liang, Elliot Meyerson, Babak Hodjat, et al.*, GECCO 2019.

- Evolutionary optimization of novel neural network architectures for multi-task learning.

Evolutionary Architecture Search for Deep Multitask Networks (2018) *Jason Liang, Elliot Meyerson, and Risto Miikkulainen*, GECCO 2018.

- Extends CoDeepNEAT to multitask architecture search, automatically discovering deep multitask networks that outperform human-designed baselines.

Evolving Deep Neural Networks (2017/2019) *Risto Miikkulainen, Jason Liang, Elliot Meyerson, et al.*, Artificial Intelligence in the Age of Neural Networks and Brain Computing.

- Book chapter surveying neuroevolutionary methods for deep learning; introduces CoDeepNEAT, the coevolutionary neural architecture search algorithm I developed, with applications to language modeling, image captioning, and multi-task learning.

Evolutionary Bilevel Optimization for Complex Control Tasks (2015) *Jason Zhi Liang, Risto Miikkulainen*, GECCO 2015.

- Bilevel evolutionary optimization of neuroevolution hyperparameters, applied to continuous-control tasks (helicopter hovering, double pole balancing).

Patents (Issued, U.S.):

- **US Patent 12,282,845:** Multi-objective Coevolution of Deep Neural Network Architectures (2025).
- **US Patent 11,507,844:** Asynchronous Evaluation Strategy for Evolution of Deep Neural Networks (2022).
- **US Patent 11,250,328:** Cooperative Evolution of Deep Neural Network Structures (2022).
- **US Patent 11,030,529:** Evolution of Architectures for Multitask Neural Networks (2021).
- **US Patent 11,003,994:** Evolutionary Architectures for Evolution of Deep Neural Networks (2021).

Patents (Pending, U.S.):

- **App. 17/389,961:** Regularized Evolutionary Population-Based Training (2023).
- **App. 17/064,706:** Sharing Meta-Learning Methods Among Multiple Private Data Sets (2022).

Awards & Honors

- **BEACON Research Grant** (2015), BEACON Center for the Study of Evolution in Action (an NSF Science & Technology Center).

Professional Activities

- **ICML Silver Reviewer**, International Conference on Machine Learning (ICML)
- **Reviewer**, Genetic and Evolutionary Computation Conference (GECCO)

Education

Ph.D. in Computer Science · *Neuroevolution & Deep Learning*
The University of Texas at Austin · Sep 2013 – Dec 2018 · Advisor: Risto Miikkulainen
Dissertation: *Evolutionary Neural Architecture Search for Deep Learning*

B.S. in Electrical Engineering and Computer Science
University of California, Berkeley · Sep 2009 – May 2013